

Mohammad Yousefi

Artificial Intelligence and Data Science Researcher

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Research Profile

Artificial Intelligence and Data Science researcher with an M.Sc. in Artificial Intelligence from Bahcesehir University and research experience in deep learning, computer vision, radar-based human detection, medical imaging, and machine learning. My work focuses on developing practical and computationally efficient AI systems for real-world applications, with particular interest in human sensing, healthcare, anomaly detection, and edge AI. I have published and presented research at IEEE conferences and have experience developing, evaluating, and optimizing deep learning models for resource-constrained environments.

Research Interests

Deep Learning; Computer Vision; Machine Learning; Medical Imaging; Radar-Based Human Detection; UWB Radar Sensing; Edge AI; Multi-modal Sensor Fusion; Human Activity and Vital-Sign Detection; Anomaly Detection

Education

M.Sc. in Artificial Intelligence

Bahcesehir University

Oct. 2022 – Nov. 2024

CGPA: 3.71 / 4.00

B.Sc. in Computer Engineering

University of Guilan

Sep. 2017 – Aug. 2021

CGPA: 16.84 / 20.00 (3.44 / 4.00)

Publications

- **Yousefi, Mohammad**, and Berjin, Emine. "A Lightweight Physics-Informed Deep Learning Framework for Human Presence Detection Using UWB Radar" *Electronics*, 2026. <https://doi.org/10.3390/electronics15184301>
- **Yousefi, Mohammad**, and Berjin, Emine. "Physics-Aware Lightweight Deep Learning Model for Human Breathing Detection via UWB Radar" *2026 34th Signal Processing and Communications Applications Conference (SIU)*, 2026. <https://doi.org/10.1109/SIU71813.2026.11636904>
- **Yousefi, Mohammad**, and Berjin, Emine. "Lightweight Edge Model for Real-Time Breathing-Based Human Presence Detection with UWB Radar" *Proceedings of the 27th International Radar Symposium (IRS)*, 2026. <https://doi.org/10.23919/IRS70539.2026.11548971>
- **Yousefi, Mohammad**, and Berjin, Emine. "Improving the Robustness of CNN-Based Human Detection in Cluttered Settings Using Multiple Raw UWB Radar Datasets" *Proceedings of the International Humanitarian Technology Conference (IHTC)*, 2025. <https://doi.org/10.1109/IHTC61819.2024.10855134>
- Berjin, Emine, and **Yousefi, Mohammad**. "Neural Network-Based Human Detection Using Raw UWB Radar Data." *Proceedings of the Microwave Theory and Technology in Wireless Communications (MTTW)*, 2024. <https://doi.org/10.1109/MTTW64344.2024.10742175>

Research Projects

UWB Radar Human Presence Detection

Jan. 2026

Developed lightweight CNN-based models for human presence detection using raw ultra-wideband (UWB) radar data. Investigated robustness under cluttered conditions and developed domain-informed features for improving detection performance. Explored model compression and post-training INT8 quantization for edge deployment.

Brain Tumor MRI Segmentation

Jun. 2025

Developed a U-Net-based deep learning model for brain tumor segmentation using MRI data. Investigated attention mechanisms and combined Dice, binary cross-entropy, and edge-aware loss functions to improve segmentation performance, particularly for challenging tumor regions.

Pathology Visual Question Answering	<i>Dec. 2023</i>
Implemented a deep learning model for answering questions about pathology images and achieved a 5% performance improvement over baseline models.	
Transformer-Based Machine Translation	<i>Mar. 2025</i>
Developed a Transformer-based English–German neural machine translation model using 10% of the WMT14 dataset, achieving a BLEU score of 46.04.	
Brain Tumor MRI Classification	<i>Apr. 2024</i>
Designed a custom convolutional neural network for four-class brain tumor classification, achieving over 94% classification accuracy.	
Real-Time Emotion Detection	<i>Dec. 2023</i>
Developed CNN and VGGNet models for facial emotion recognition using FER2013, achieving 81% accuracy while reducing the parameter count by 54% compared with state-of-the-art models.	
Malware Detection Using UNSW-NB15	<i>Dec. 2023</i>
Developed and evaluated DNN, GRU, and Random Forest models for network intrusion and malware-related classification using the UNSW-NB15 dataset, achieving up to 95% accuracy.	

Technical Skills

Programming: Python, SQL, C++, Java, JavaScript
Machine Learning and Deep Learning: TensorFlow, PyTorch, Scikit-learn, OpenCV
Data Science: Pandas, NumPy, Matplotlib, TensorBoard, Jupyter Notebook
Software and Tools: MATLAB, SQL Server, Tableau, VS Code, Wireshark, Git
Operating Systems: Ubuntu, Kali Linux, Manjaro, Raspberry Pi OS
Other: LaTeX, Shell Scripting

Professional Experience

IT Support Intern	<i>Jan. 2021 – Jan. 2023</i>
<i>Pishgaman-e Guilan Export Development</i>	
- Managed computer network security and provided technical support for organizational IT infrastructure. - Developed internal software that improved operational efficiency by approximately 50%. - Prepared technical project documentation and achieved a 93% approval rate.	

Certifications

- **Deep Neural Networks with PyTorch** – IBM, Sep. 2022
- **Machine Learning** – Stanford University, Mar. 2022
- **Deep Learning Specialization** (5 courses) – DeepLearning.AI, Jul. 2022
- **Applied Data Science in Python Specialization** (5 courses) – University of Michigan, Feb. 2021

English Proficiency and Standardized Tests

IELTS Academic	<i>May 24, 2024</i>
Overall Band Score: 7.5 (Listening: 8.0, Reading: 7.5, Writing: 6.5, Speaking: 7.0)	
Graduate Record Examination (GRE)	<i>Aug. 9, 2022</i>
Quantitative: 164 Verbal: 145 Analytical Writing: 3.5	

Academic References

References available upon request.